ONLINE MASTERS IN **DATA SCIENCE**

DSC 258 – NATURAL LANGUAGE PROCESSING

TRAINING RNN LANGUAGE MODELS

UC San Diego

COMPUTER SCIENCE & ENGINEERING

HALICIOĞLU DATA SCIENCE INSTITUTE

Likelihood

Single sentence: $x = (x_1, \ldots, x_n)$

$$\log p(x_1, \ldots, x_n) = \sum_i \log p(x_i \mid x_1, \ldots, x_{i-1})$$

Corpus of sentences: $x^{(1)} = (x_1^{(1)}, \ldots, x_n^{(1)}) \quad \cdots \quad x^{(K)} = (x_1^{(K)}, \ldots, x_n^{(K)})$

$$\sum_k \log p(x_1^{(k)}, \ldots, x_n^{(k)}) = \sum_k \sum_i \log p(x_i^{(k)} \mid x_1^{(k)}, \ldots, x_{i-1}^{(k)})$$

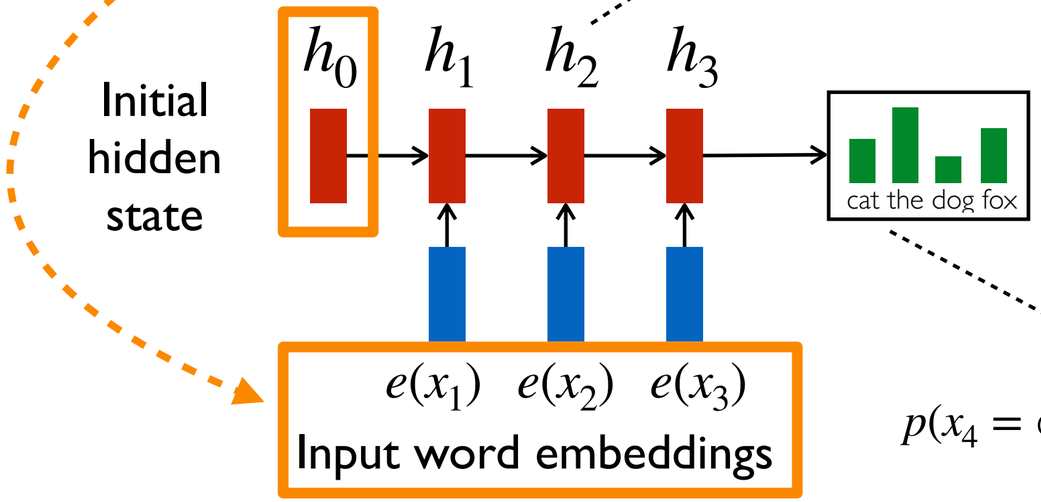
Parameters

$$p(x_1, \ldots, x_n; \theta) = \prod_i p(x_i \mid x_1, \ldots, x_{i-1}; \theta)$$

Model parameters θ

$$\theta = (E, V, U, W, h_0)$$

$$h_i = \sigma(W h_{i-1} + U e(x_i))$$



$$p(x_4 = \text{cat} \mid x_1, x_2, x_3) = \frac{\exp(v_{\text{cat}} \cdot h_3)}{\sum_x \exp(v_x \cdot h_3)}$$

Training

Training $x^{(1)} = (x_1^{(1)}, \ldots, x_n^{(1)}) \quad \cdots \quad x^{(K)} = (x_1^{(K)}, \ldots, x_n^{(K)})$

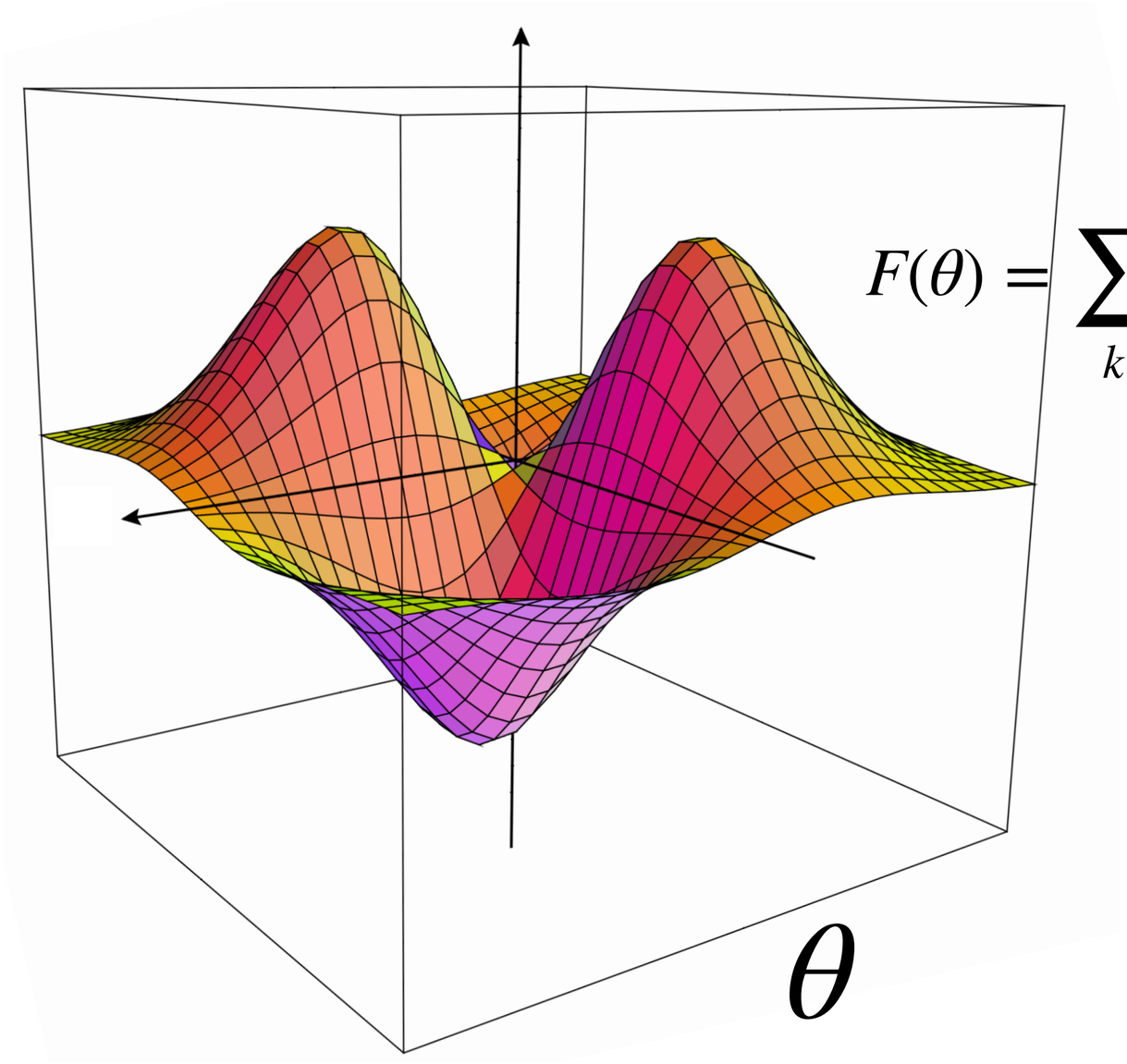
Training

$$\max_\theta \sum_k \log p(x_1^{(k)}, \ldots, x_n^{(k)}; \theta) = \max_\theta \sum_k \sum_i \log p(x_i^{(k)} \mid x_1^{(k)}, \ldots, x_{i-1}^{(k)}; \theta)$$

Follow the gradient of the

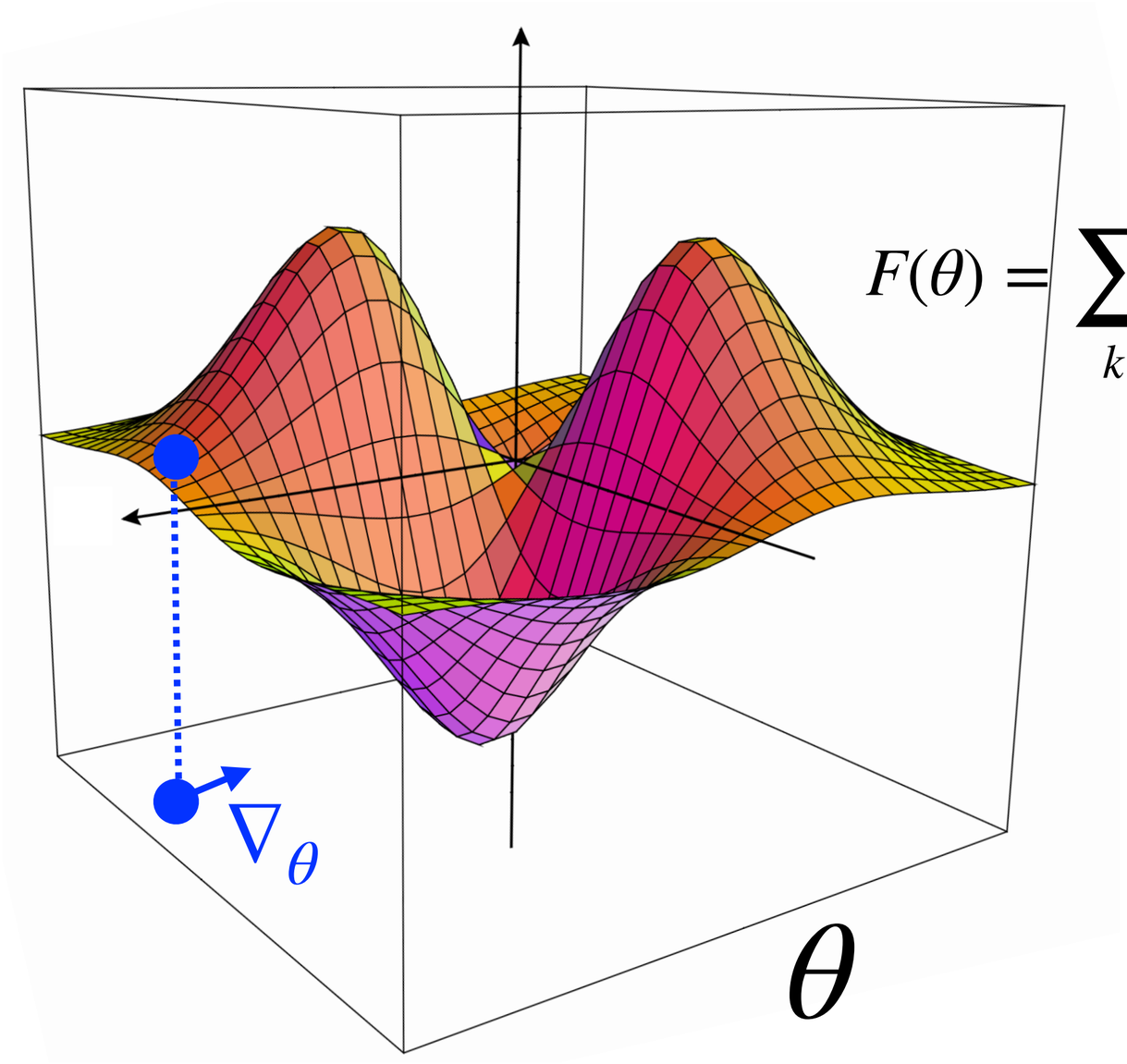
$$\theta = \theta + \eta \cdot \boxed{\nabla_\theta \sum_k \log p(x_1^{(k)}, \ldots, x_n^{(k)}; \theta)}$$

Gradient Ascent



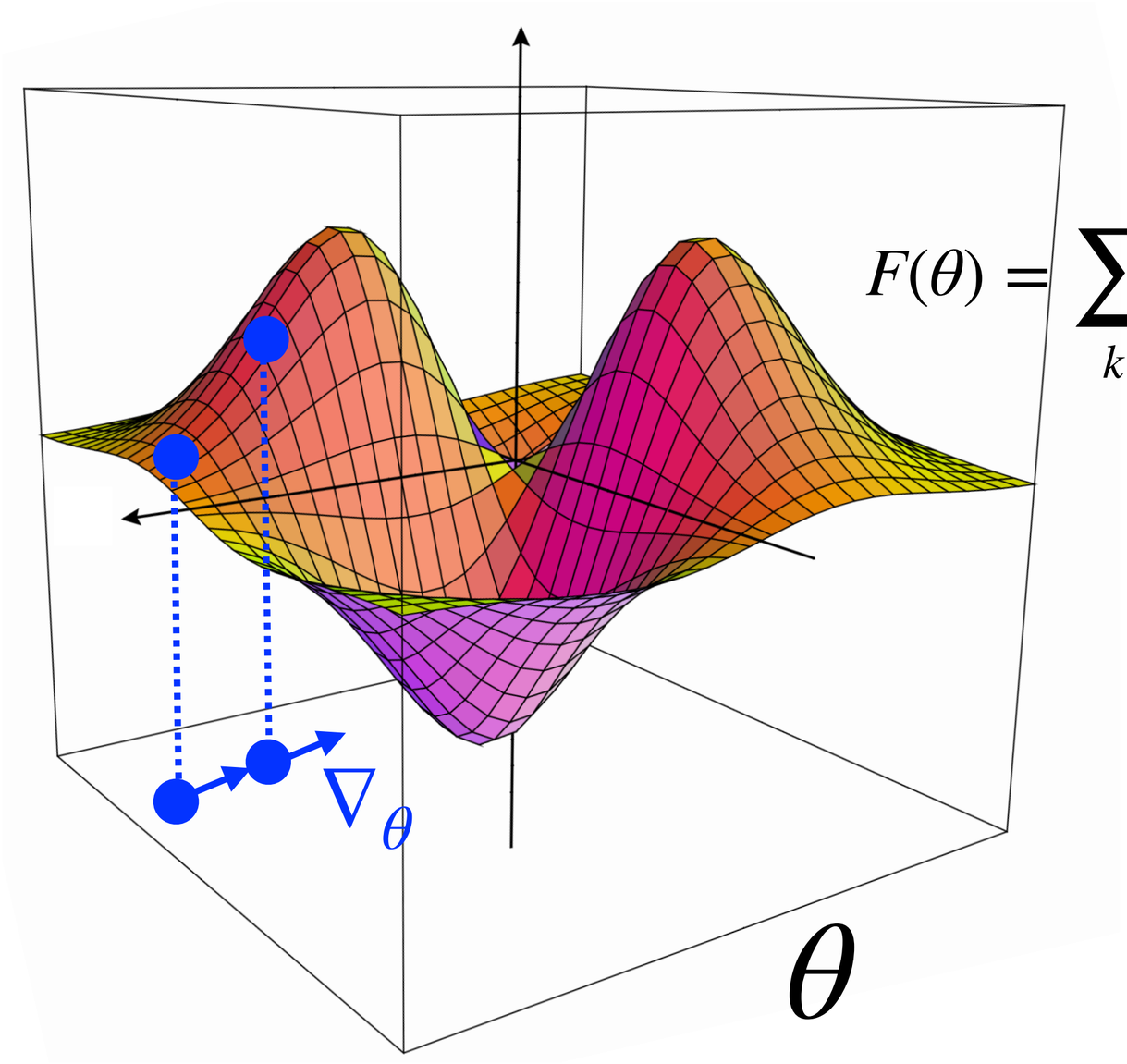
$$F(\theta) = \sum_k \log p(x_1^{(k)}, \ldots, x_n^{(k)}; \theta)$$

Gradient Ascent



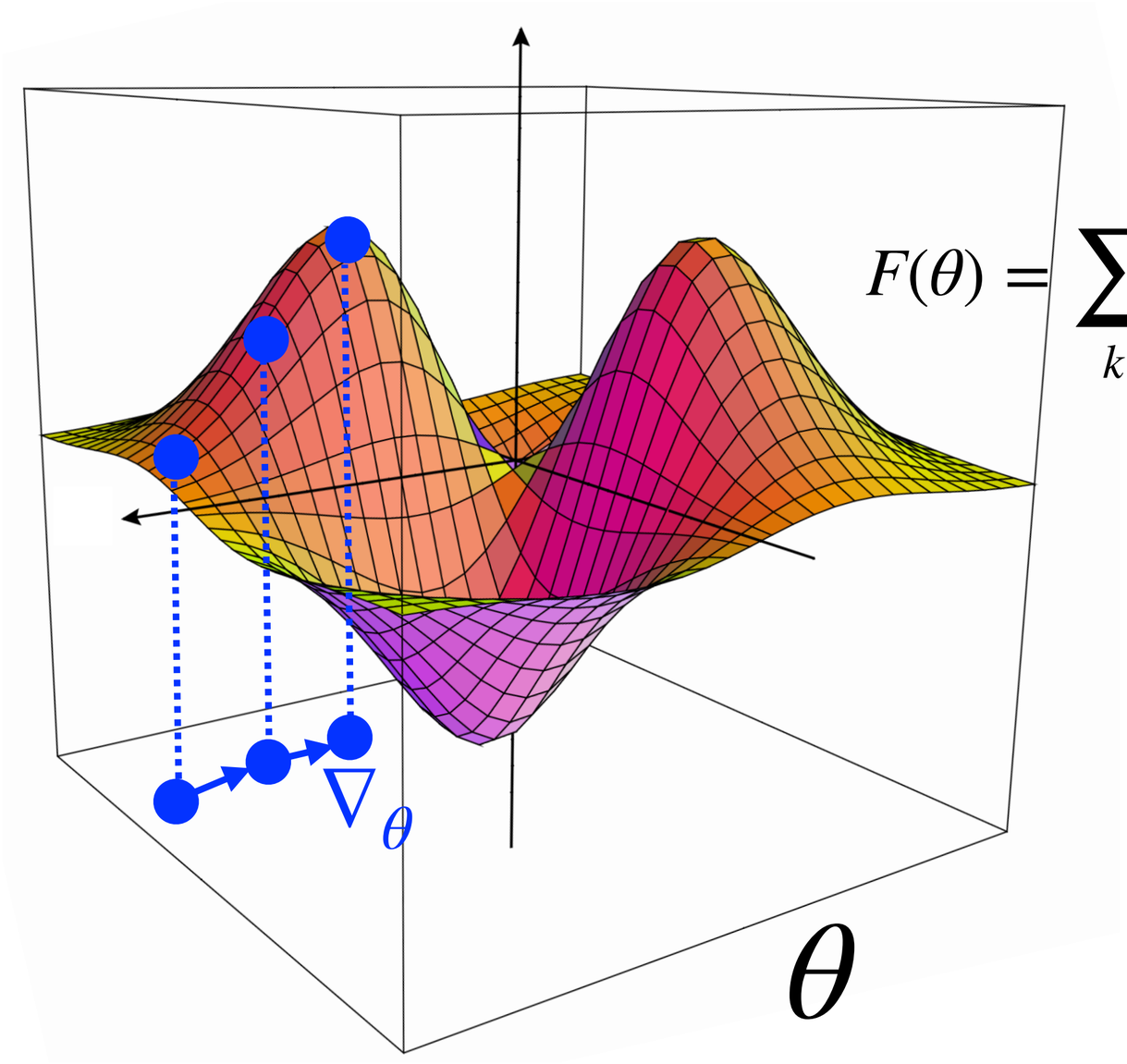
$$F(\theta) = \sum_k \log p(x_1^{(k)}, \ldots, x_n^{(k)}; \theta)$$

Gradient Ascent



$$F(\theta) = \sum_k \log p(x_1^{(k)}, \ldots, x_n^{(k)}; \theta)$$

Gradient Ascent



$$F(\theta) = \sum_k \log p(x_1^{(k)}, \ldots, x_n^{(k)}; \theta)$$